



# High Density Polyethylene

## Aramco HDPE F5101

### Product Description

Aramco HDPE F5101 is a high molecular weight high density polyethylene resin and is typically used in for blown film process especially for thin film applications. This resin has broad molecular weight distribution that makes it easier to process. Films made from this resin typically exhibits high stiffness with excellent impact and toughness characteristics.

### Typical Applications

Shopping bags, trash bags, industrial liners.

This product is not intended for use in medical and pharmaceutical applications.

| Physical Properties               |    | Unit              | Test Method        | Value <sup>(1)</sup> |
|-----------------------------------|----|-------------------|--------------------|----------------------|
| Melt Flow Index (190°C/2.16kg)    |    | g/10min           | ASTM D1238         | 0.05                 |
| Melt Flow Index (190°C/21.6kg)    |    | g/10min           | ASTM D1238         | 9.3                  |
| Density                           |    | g/cm <sup>3</sup> | ASTM D792 Method A | 0.951                |
| Film Properties <sup>(2)</sup>    |    |                   |                    |                      |
| Tensile Strength at Yield         | MD | MPa               | ASTM D882          | 35                   |
|                                   | TD |                   |                    | 30                   |
| Tensile Strength at Break         | MD | MPa               | ASTM D882          | 75                   |
|                                   | TD |                   |                    | 55                   |
| Tensile Elongation at Break       | MD | %                 | ASTM D882          | 520                  |
|                                   | TD |                   |                    | 560                  |
| 1% Secant Modulus                 | MD | MPa               | ASTM D882          | 990                  |
|                                   | TD |                   |                    | 1140                 |
| Elmendorf Tear Strength           | MD | gf                | ASTM D1922         | 11                   |
|                                   | TD |                   |                    | 78                   |
| Dart Drop Impact                  |    | f                 | ASTM D1709         | 245                  |
| Thermal Properties <sup>(2)</sup> |    |                   |                    |                      |
| Vicat Softening Temperature @10N  |    | °C                | ASTM D1525         | 126                  |

(1) Typical values only, not to be construed as specification limits

(2) Film fabrication conditions: Extruder 50mmφ, Die 120mmφ, Die Gap 1.35mm, Temp 200°C, Output 50kg/hr., BUR 4.0, Thickness 20μm, Neck Height 840mm

**Processing conditions:** Recommended melt temperature: 180 – 220 °C

**Storage and handling:** Aramco HDPE F5101 should be stored in a dry cool place with adequate ventilation and protected from UV-light at temperature below 50 °C

**Food Contact and other Regulations:** Aramco HDPE F5101 resins are suitable for food contact applications. Detailed information is provided in Material Safety Datasheet and for additional specific information, please contact Aramco Trading for certificates.

Disclaimer: © Saudi Aramco Products Trading Company. All information supplied by or on behalf Saudi Aramco Products Trading Company in relation to its products, whether in the nature of data, recommendations or otherwise, is supported by research and believed reliable, but Saudi Aramco Products Trading Company assumes no liability whatsoever in respect of application, processing or use made of the provided information or products, or any consequence thereof. The user is solely responsible and liable with respect to the application, processing or use of the provided information or product, whose quality and other properties shall be verified, or any consequence thereof. No liability whatsoever shall attach Saudi Aramco Products Trading Company for any infringement of the rights owned or controlled by a third party in intellectual, industrial or other property by reason of the application, processing or use of the provided information or products by the user. To the extent the user is entitled to disclose and distribute this document, the user may forward, distribute, and/or photocopy this copyrighted document only if unaltered and complete, including all of its headers, footers, disclaimers and other information.